

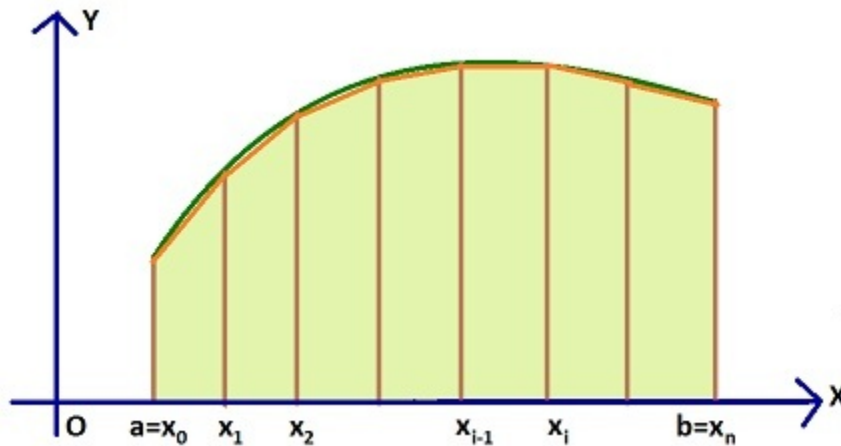
Note0

I tried not to be academic and not to take much of your time so forgive me if i made a mistake.

Trapezoid Rule

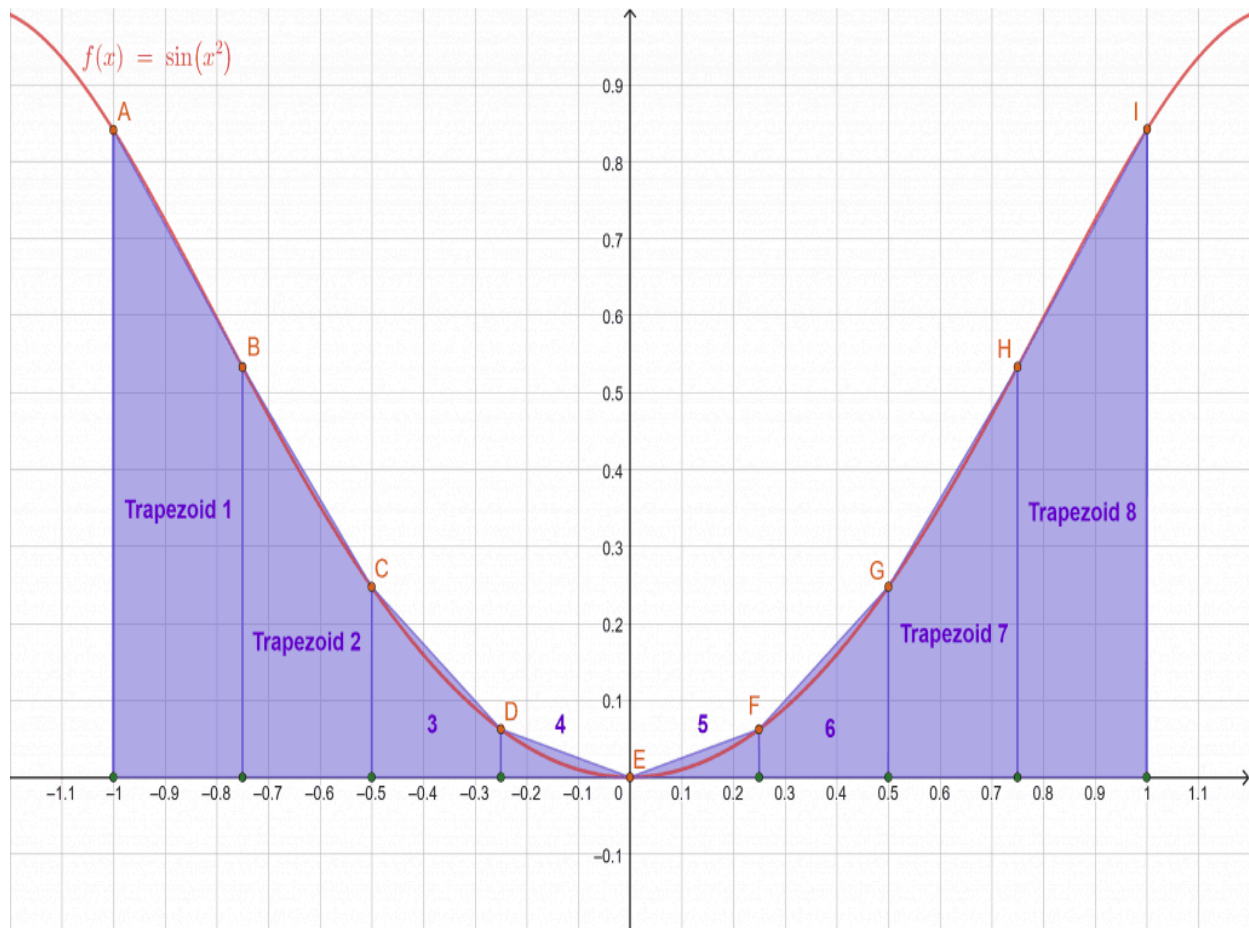
What is the Trapezoidal Rule?

We use the trapezoid rule to estimate the value of the definite integral which represents the area under the curve.



How to Use the Trapezoidal Rule?

We calculate the whole area under the curve by creating sub-intervals to make it easy to calculate areas and adding all those areas together.



Note1:

The general rule of the sum of Areas :

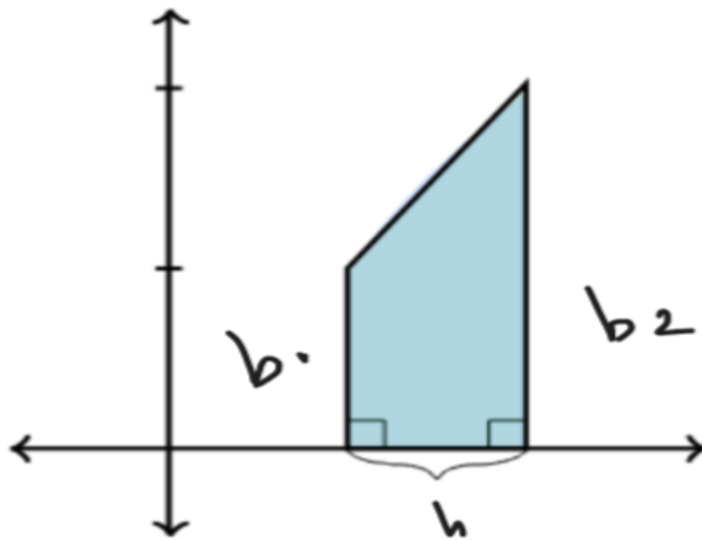
$$S_{\text{Trap}} = \sum_{i=1}^n \frac{f(x_{i-1}) + f(x_i)}{2} \cdot \Delta x$$

n : The Number of the subintervals , The more trapezoids we take the more accurate the integration (Area).

Note2:

The Area of trapezoid:

$$V_{1j} = \frac{h_1}{2} (f(a) + f(b))$$



deduction :

The Sum of number of areas under the curve between the number of intervals = The Definite integral to the number of intervals.

Example:

$$\int_0^1 x^2 dx \quad \text{using } n = 4$$

$$h = \frac{1-0}{4} = 0.25$$

4 values of $x = 0, 0.25, 0.50, 0.75, 1$

Applying the formula:

$$0.5 \cdot 0.25 [0 + 2(0.0625 + 0.25 + 0.5625) + 1]$$

=

$$\boxed{0.34375}$$

m^2

$$\int_0^1 x^2 \quad \downarrow x$$

$$\frac{1}{3} \left[x^3 \right]_0^1 = \frac{1}{3}$$

$$1/3 =$$

$$\boxed{0.3333}$$

m^2

- Sources and references:

1. https://youtu.be/XMjlr_7IUI0?si=bFF13b0dMSvONRkB
2. <https://youtu.be/Rn9Gr52zhrY?si=JdeTcyBlmb0wVPLZ>